

Machine Learning for Personalized Music Recommendation Based on User Listening Preferences

1. Title

Machine Learning for Personalized Music Recommendation Based on User Listening Preferences

2. Abstract

Streaming platforms have made music libraries too large to browse manually. This project uses machine learning to generate personalized recommendations from observed listening behaviour, rather than stated preferences.

We will select a public dataset, define a concrete recommendation task, and build a simple baseline plus at least two substantive ML models, evaluated with a consistent train/validation/test split using ranking metrics (Precision@K, Recall@K, NDCG@K).

The result will be a reproducible proof-of-concept, including analysis of failure modes such as sparse listening histories and popularity bias.

3. Research Background

Large streaming libraries make music discovery difficult. Recommendation systems address this by surfacing items likely to interest a user based on past behaviour.

This project uses observed listening interactions rather than explicit ratings — a signal we can learn from without assuming every interaction reflects a genuine preference.

Research question: How effectively can machine learning use observed listening behaviour to generate personalized recommendations?

We will explore the chosen dataset first — user and song counts, interaction sparsity, and distribution — before committing to a specific task or method, since the right approach depends on the data. Known challenges include sparse listening histories, popularity bias, and cold-start users or songs; no single technique is assumed in advance to be the right fit.

4. Methodology

Data

We will use a public dataset of user-song listening interactions, selected during an initial review for relevance, size, documentation, and licensing. Preprocessing will include removing invalid or duplicate records, handling missing values, and reshaping the data for modelling.

Methods

We will implement a simple baseline and at least two substantive ML approaches — for example, user- or item-similarity methods — chosen once the dataset and task are finalized. All approaches will share the same preprocessing and evaluation pipeline, and we will document the reasoning behind each method.

Experiments and Evaluation

The dataset will be split into training, validation, and test sets. Validation data will guide model selection and tuning; the test set is reserved for final evaluation. We will log preprocessing steps, splits, seeds, and parameters for reproducibility, and analyze errors — for example, how performance varies with listening-history length or exposure to popular songs.

Evaluation Metrics

Since this is a ranking problem, we will use Precision@K, Recall@K, and NDCG@K, selecting a primary metric once the task is finalized.

Comparison

All approaches, including the baseline, will be evaluated on identical data splits and metrics.

5. Significance

This project will produce a reproducible proof-of-concept for applying machine learning to implicit listening-behaviour data — useful for others working on recommendation systems with sparse or implicit feedback. The controlled comparison will also show which data characteristics drive performance and where methods fail, providing a foundation for future work incorporating song, artist, or genre metadata.

6. SMART Timeline

Sprint	Time Period	Objectives	Deliverable
1 — Problem & Data Investigation	Sept. 18 – Oct. 2	Explore candidate datasets, assess available user/song data, and define the research question and recommendation problem.	Selected dataset, research question, initial data analysis
2 — Task & Baseline	Oct. 3 – Oct. 19	Define the recommendation task, set the evaluation methodology and metrics, and implement a baseline.	Evaluation procedure, baseline implementation, initial results
3 — ML Approach Investigation	Oct. 20 – Oct. 30	Select and implement at least two ML approaches; begin validation experiments.	Working ML approaches, initial comparison
4 — Final Experiments & Analysis	Oct. 31 – Nov. 20	Tune models, run controlled experiments against the baseline, and analyze errors and limitations.	Final results, comparison, error analysis
5 — Finalization & Presentation	Nov. 21 – Dec. 4	Finalize the pipeline, documentation, report, and presentation; prepare for oral defense.	Final repository, report, presentation, demonstration

7. References

Final academic references and dataset documentation will be added once the team selects and verifies its data sources, using APA or IEEE style per the proposal template.